

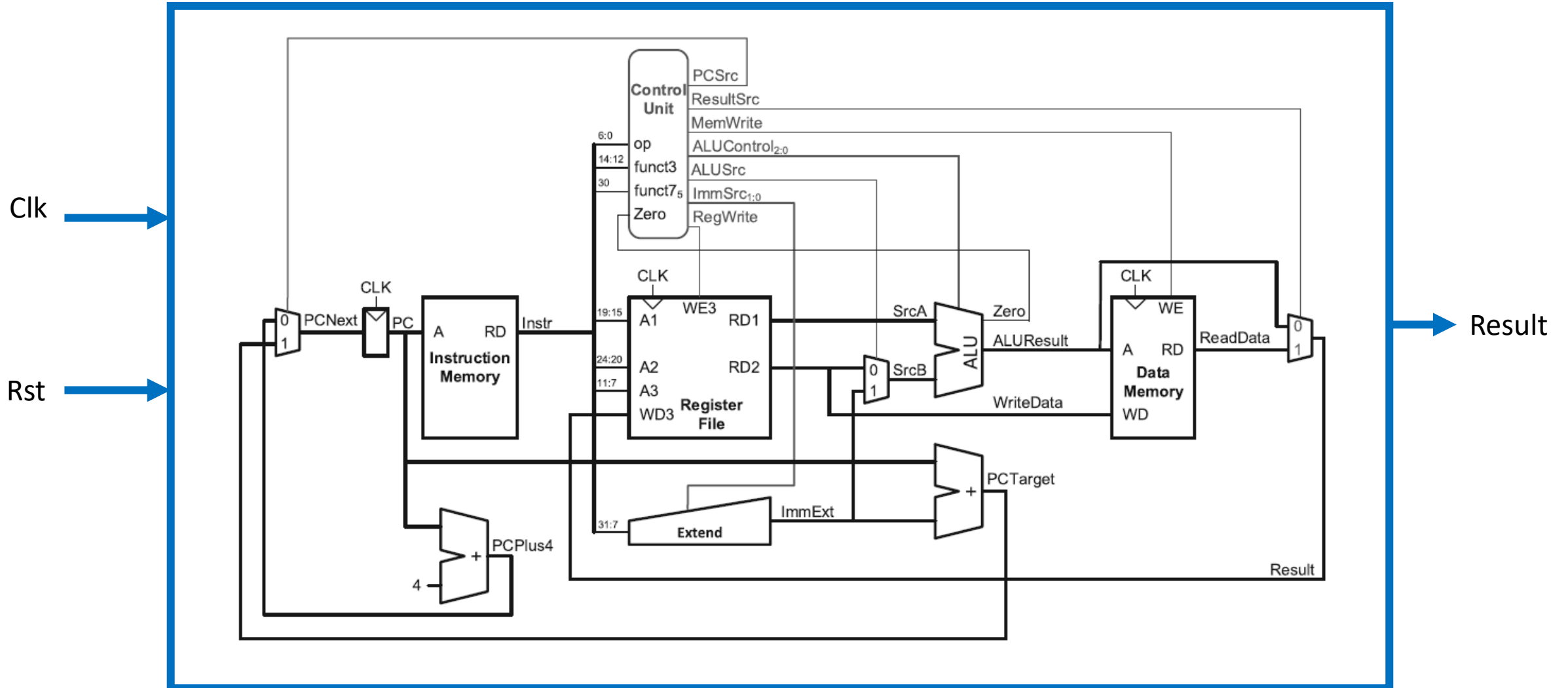
Digital System Design

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Fall 2026 · Lecture 7



RISC_V Microarchitecture



Microarchitecture

- Microarchitecture connects the instruction set architecture to the digital hardware.
- It describes how registers, memories, ALUs, multiplexers, and control logic work together.
- **We want to develop a single-cycle RISC-V processor.**

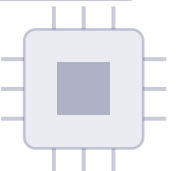
Application software

Operating system

Architecture

Microarchitecture

Digital circuits



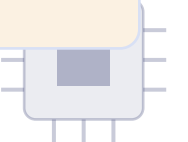
Architecture and Microarchitecture

1 Computer Architecture

- Defines the programmer's view of the processor.
- Specifies the instruction set, registers, data types, and memory behavior.
- Describes what the processor must do.

2 Microarchitecture

- Defines the internal hardware used to implement the architecture.
- Describes how the processor performs each instruction.
- Different microarchitectures can implement the same architecture.

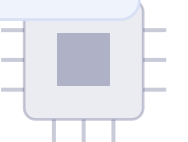


Architectural State

- Contains the information required to describe the current state of a program.
- For the RV32I processor, it includes:
 - The 32-bit program counter, PC
 - Thirty-two 32-bit registers, x0 through x31
 - The contents of memory

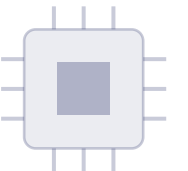
State transition

The processor uses the current architectural state and instruction to produce the next architectural state.



Supported RISC-V Instructions

Category	Instructions
R-type ALU	add, sub, and, or, slt
Memory	lw, sw
Branch	beq
Added later	addi, andi, ori, slti, jal



Datapath and Control Unit

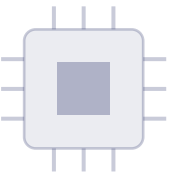
Control Unit

- Examines the current instruction
- Selects the required datapath operation
- Generates mux selects, write enables, and ALU control signals

Control
signals

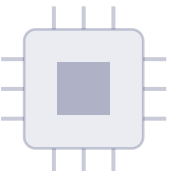
Datapath

- Operates on data values
- Contains registers, memories, ALUs, adders, and multiplexers
- Uses 32-bit data paths for RV32I



Processor Design Process

- 1 Identify the architectural state elements.
- 2 Connect combinational logic between the state elements.
- 3 Determine how each instruction moves through the datapath.
- 4 Add multiplexers when instructions require different data sources.
- 5 Generate the control signals required to select the correct paths.
- 6 Analyze the critical path and execution time.



State Elements of a RISC-V Processor

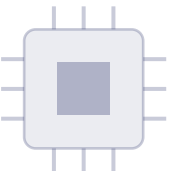
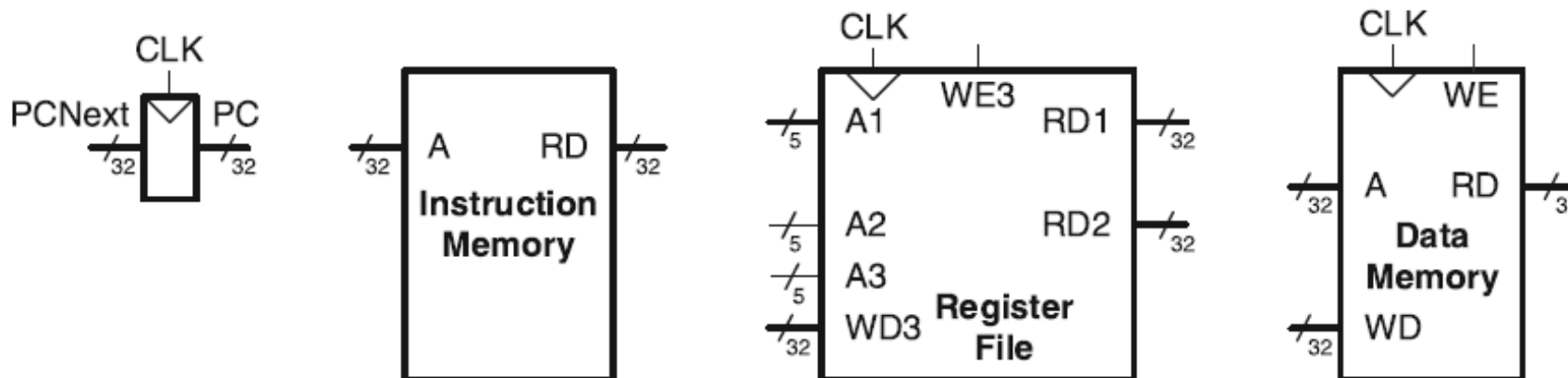
- The initial processor contains four state elements:
 - Program counter
 - Instruction memory
 - Register file
 - Data memory
- The program counter holds the address of the current instruction.

Register file

Two combinational read ports

One clocked write port

Register x0 always contains zero



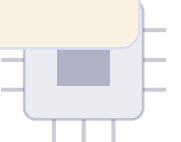
Reading and Writing State Elements

1 Combinational Reads

- Instruction memory, data memory, and the register file produce new outputs when their addresses change.
- Reading does not require a clock edge.

2 Synchronous Writes

- The PC, register file, and data memory update on the rising clock edge.
- Input data and control signals must satisfy setup and hold time requirements.



Single-Cycle Processor

- A single-cycle processor completes every instruction in one clock cycle.
- The control unit uses combinational logic.
- No intermediate state registers are required.
- Every instruction uses the same clock period.
- The slowest instruction determines the clock period.
- Separate instruction and data memories allow simultaneous access.

CPI

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Every instruction — fast or slow — takes exactly one clock cycle.

